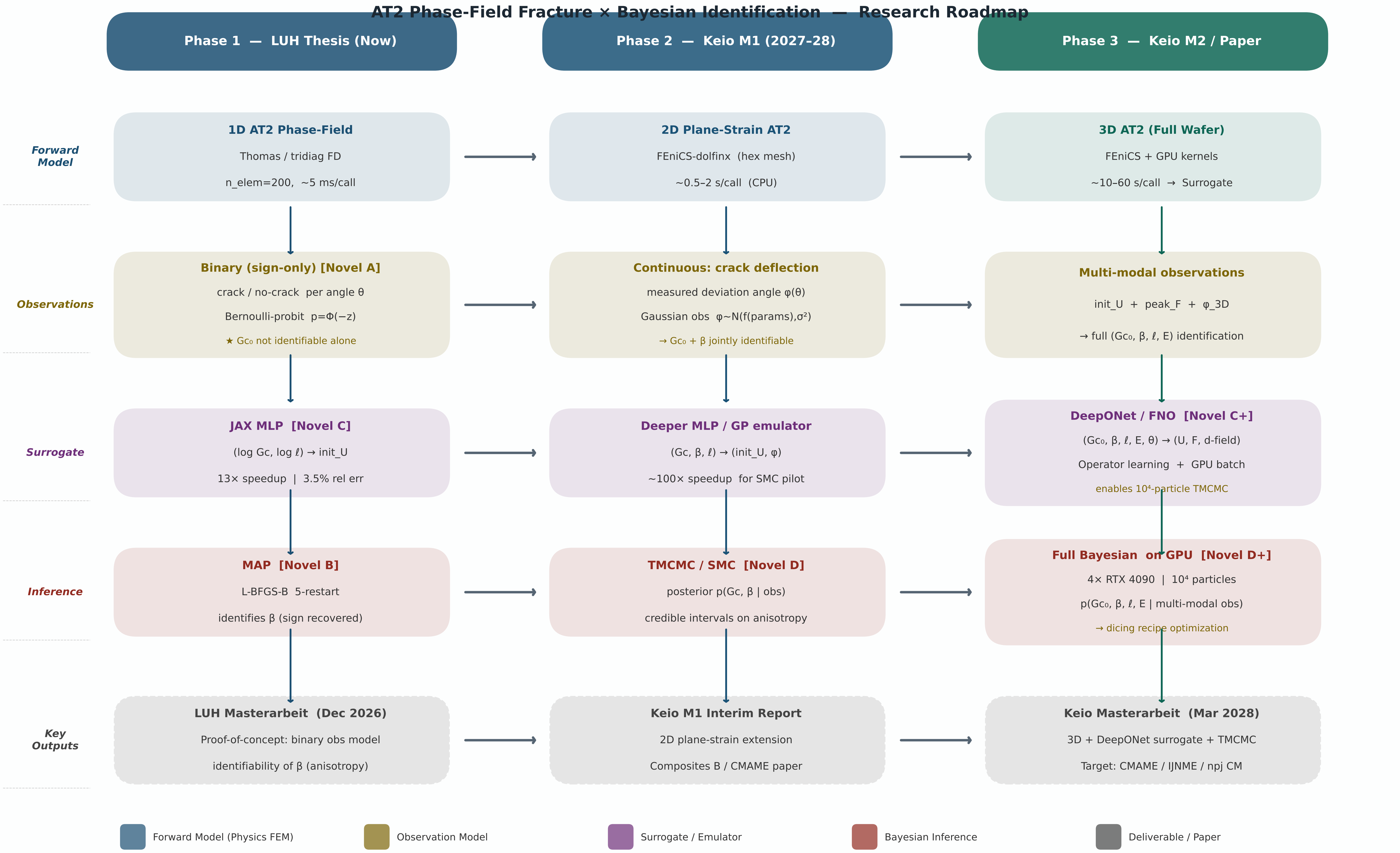
AT2 Phase-Field Fracture × Bayesian Identification — Research Roadmap
Phase 1 — LUH Thesis (Now)
Phase 2 — Keio M1 (2027–28)
Phase 3 — Keio M2 / Paper
Forward Model
1D AT2 Phase-Field
Thomas / tridiag FD
n_elem=200, ~5 ms/call
2D Plane-Strain AT2
FEniCS-dolfinx (hex mesh)
~0.5–2 s/call (CPU)
3D AT2 (Full Wafer)
FEniCS + GPU kernels
~10–60 s/call → Surrogate
Observations
Binary (sign-only) [Novel A]
crack / no-crack per angle θ
Bernoulli-probit p=Φ(−z)
★ Gc_0 not identifiable alone
Continuous: crack deflection
measured deviation angle φ(θ)
Gaussian obs φ~N(f(params),σ^2)
→ Gc_0 + β jointly identifiable
Multi-modal observations
init_U + peak_F + φ_3D
→ full (Gc_0, β, ℓ, E) identification
Surrogate
JAX MLP [Novel C]
(log Gc, log ℓ) → init_U
13× speedup | 3.5% rel err
Deeper MLP / GP emulator
(Gc, β, ℓ) → (init_U, φ)
~100× speedup for SMC pilot
DeepONet / FNO [Novel C+]
(Gc_0, β, ℓ, E, θ) → (U, F, d-field)
Operator learning + GPU batch
enables 10^4-particle TMCMC
Inference
MAP [Novel B]
L-BFGS-B 5-restart
identifies β (sign recovered)
TMCMC / SMC [Novel D]
posterior p(Gc, β | obs)
credible intervals on anisotropy
Full Bayesian on GPU [Novel D+]
4× RTX 4090 | 10^4 particles
p(Gc_0, β, ℓ, E | multi-modal obs)
→ dicing recipe optimization
Key Outputs
LUH Masterarbeit (Dec 2026)
Proof-of-concept: binary obs model
identifiability of β (anisotropy)
Keio M1 Interim Report
2D plane-strain extension
Composites B / CMAME paper
Keio Masterarbeit (Mar 2028)
3D + DeepONet surrogate + TMCMC
Target: CMAME / IJNME / npj CM
Forward Model (Physics FEM)
Observation Model
Surrogate / Emulator
Bayesian Inference
Deliverable / Paper